- vector equation of the plane in parametric form is $\vec{r}=\vec{a}+s\left(\vec{b}-\vec{a}\right)+t\vec{c}$, where $s,\ t\in\mathbb{R}$ which implies that $\vec{r}=\left(-\hat{i}+2\hat{j}\right)+s\left(3\hat{i}-\hat{k}\right)+t\left(\hat{i}+\hat{j}-\hat{k}\right)$, where $s,\ t\in\mathbb{R}$.
- vector equation of the plane in non-parametric form is $(\vec{r}-\vec{a})\cdot((\vec{b}-\vec{a})\times\vec{c})=0$.

  Now, $(\vec{b}-\vec{a})\times\vec{c}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\3&0&-1\\1&1&-1\end{vmatrix}=\hat{i}+2\hat{j}+3\hat{k}$,

  we have $(\vec{r}-(-\hat{i}+2\hat{j}))\cdot(\hat{i}+2\hat{j}+3\hat{k})=0\Rightarrow\vec{r}\cdot(\hat{i}+2\hat{j}+3\hat{k})=3$
- If $\vec{r}=x\hat{i}+y\hat{j}+z\hat{k}$ is the position vector of an arbitrary point on the plane, then from the above equation, we get the Cartesian equation of the plane as $x+2y+3z=3$.

EXERCISE 6.7

1. Find the non-parametric form of vector equation, and Cartesian equation of the plane passing through the point $(2,3,6)$ and parallel to the straight lines $\frac{x-1}{2}=\frac{y+1}{3}=\frac{z-3}{1}$ and $\frac{x+3}{2}=\frac{y-3}{-5}=\frac{z+1}{-3}$
2. Find the non-parametric form of vector equation, and Cartesian equations of the plane passing through the points $(2,2,1)$, $(9,3,6)$ and perpendicular to the plane $2x+6y+6z=9$.
3. Find parametric form of vector equation and Cartesian equations of the plane passing through the points $(2,2,1),(1,-2,3)$ and parallel to the straight line passing through the points $(2,1,-3)$ and $(-1,5,-8)$.
4. Find the non-parametric form of vector equation and cartesian equation of the plane passing through the point $(1,-2,4)$ and perpendicular to the plane $x+2y-3z=11$ and parallel to the line $\frac{x+7}{3}=\frac{y+3}{-1}=\frac{z}{1}$.
5. Find the parametric form of vector equation, and Cartesian equations of the plane containing the line $\vec{r}=(\hat{i}-\hat{j}+3\hat{k})+t(2\hat{i}-\hat{j}+4\hat{k})$ and perpendicular to plane $\vec{r}\cdot(\hat{i}+2\hat{j}+\hat{k})=8$.
6. Find the parametric vector, non-parametric vector and Cartesian form of the equations of the plane passing through the three non-collinear points $(3,6,-2),(-1,-2,6)$, and (6, 4, –2).
7. Find the non-parametric form of vector equation, and Cartesian equations of the plane $\vec{r}=\left(6\hat{i}-\hat{j}+\hat{k}\right)+s\left(-\hat{i}+2\hat{j}+\hat{k}\right)+t\left(-5\hat{i}-4\hat{j}-5\hat{k}\right)$.

6.8.7 Condition for a line to lie in a plane

We observe that a straight line will lie in a plane if every point on the line, lie in the plane and the normal to the plane is perpendicular to the line.

(i) If the line $\vec{r}=\vec{a}+t\vec{b}$ lies in the plane $\vec{r}\cdot\vec{n}=d$, then $\vec{a}\cdot\vec{n}=d$ and $\vec{b}\cdot\vec{n}=0$.

(ii) If the line $\frac{x-x_1}{a}=\frac{y-y_1}{b}=\frac{z-z_1}{c}$ lies in the plane $Ax+By+Cz+D=0$, then $Ax_1+By_1+Cz_1+D=0$ and $aA+bB+cC=0$.

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